"We are software engineering students. We designed a process for SmartDose
GmbH, a company adding an ML dosage-recommendation model to a smart insulin
pen. These were our decisions: You have budget for exactly 2 new hires.
You need all four of these roles:
(1) ML Engineer - builds and trains the model
(2) Clinical Validator - tests whether the model is medically safe
(3) Data Privacy Officer - ensures GDPR compliance for

Mehr anzeigen

Your choices are generally defensible, but the strongest critique comes from looking at what could go wrong if your assumptions fail. Here's how I would challenge each decision.

1. Hiring the ML Engineer and Clinical Validator first

Your choice

- Hire: ML Engineer + Clinical Validator
- Delay: Data Privacy Officer + Regulatory Specialist

Strongest argument for hiring the Data Privacy Officer instead of the ML Engineer

Without legally usable data, the ML Engineer cannot do meaningful work.

If patient data later turns out to be unusable under GDPR, all model development based on that data may need to be discarded. A Data Privacy Officer helps ensure that:

- Data collection is lawful from day one.
- Consent and processing grounds are documented.
- Data-sharing agreements are valid.
- Future audits do not invalidate months of work.

In a regulated medical-AI project, building the model is not usually the longest risk; building a model on data you are not allowed to use can destroy the entire project.

Strongest argument for hiring the Regulatory Specialist instead of the Clinical Validator

Medical-device regulation influences system design from the beginning.

The Regulatory Specialist would:

- Define evidence requirements.
- Set up documentation processes.
- Ensure traceability and risk-management records are created from day one.
- Prevent expensive redesigns later.

A common problem in medical-device projects is building something technically impressive that cannot be approved because required documentation was never generated during development.

Challenge to your bottleneck assessment

You said:

- Privacy becomes critical at first field testing.
- Regulatory becomes critical at release.

The strongest counterargument is that both become bottlenecks much earlier.

Data Privacy Officer
Critical as soon as patient data is acquired.

If the legal basis for using Hospital A's data is unclear, every day spent training the model increases project risk.

Regulatory Specialist
Critical during requirements and architecture design.

Under the EU Medical Device Regulation (MDR), evidence generation, risk management, documentation, and quality processes start very early. Waiting until release may mean discovering that required evidence was never collected.

2. Choosing Hospital A's dataset

Your choice

Hospital A (50,000 records, no explicit AI consent)

Strongest argument for choosing Hospital B instead

Patient safety is not the only safety issue—legal and ethical safety matters too.

Hospital B offers:

- Clear informed consent.
- Lower GDPR risk.
- Stronger public trust.
- Lower chance of project cancellation.

A smaller but legally usable dataset may be preferable to a larger dataset that regulators later prohibit.

Strongest argument for choosing neither

The company could first:

- Obtain new consent,
- Negotiate additional data-sharing agreements,
- Collect prospective data,
- Or combine multiple compliant datasets.

This delays development but avoids both:

- Unsafe under-trained models,
- And potentially unlawful data use.

Consequences you may be underestimating

You focused on retraining if Hospital A becomes unusable.

A stronger concern is that:

- Regulators could question the entire development process.
- Investors may lose confidence.
- Patients may lose trust.
- The company could face investigations or fines.
- Clinical evidence generated using the data may become questionable.

In other words, the risk is not only "retrain the model"; it could be "repeat large parts of the project."

3. Choosing Shadow Mode

Your choice

Option 2: Shadow Mode

This is probably the strongest choice overall because it balances legal compliance and progress.

Strongest argument for Option 1 (push back to month 18)

It eliminates almost all regulatory ambiguity.

Benefits:

- No chance of accidental patient exposure.
- Clear compliance story.
- Simpler risk management.
- Lower legal liability.

A medical-device company that violates regulations can suffer consequences far more severe than losing a few investors.

The argument is essentially:

> "A six-month demo is not worth risking the entire company."

Strongest argument for Option 3 (launch outside the EU first)

Real-world data is often more valuable than simulated testing.

Benefits:

- Faster model improvement.
- More edge cases discovered.
- Potentially stronger evidence package.
- Earlier market validation.

The CEO could argue that:

- The product helps patients sooner.
- Revenue starts earlier.
- The company gains practical experience before EU launch.

Challenge to your timeline

You proposed:

- Month 0–6: build shadow prototype.
- Month 6–18: doctors review outputs.
- Month 18: release.

The strongest criticism is that MDR approval is not something that automatically finishes at month 18.

More realistic:

Month	Activity
0–6	Build prototype and quality-management processes
6–18	Shadow mode studies, physician review, evidence collection
12–24+	Regulatory submission and regulator review
18–30+	Questions, additional evidence requests, revisions
After approval	Patient-facing deployment

Many medical-device approvals take longer than the theoretical minimum.

Overall assessment

The strongest critique of your entire strategy is:

> You prioritized technical feasibility and clinical performance before legal and regulatory feasibility.

This is exactly how most software projects are run, but medical AI is unusual because a technically excellent model can still fail if:

- the data was collected improperly,
- required documentation is missing,
- approval requirements were ignored early,
- or regulatory evidence was not generated from the start.

A regulator would likely challenge your choices more aggressively than an engineer would. An engineer tends to ask, "Can we build it?" while regulators ask, "Can we legally and safely deploy it?" In medical AI, the second question often becomes the harder one.